

inflows) constitutes the primary source of supply of foreign exchange. The market value of the exchange rate is determined by demand and supply in the foreign exchange market. However, often the official exchange rate may differ from its market value as you will see from our discussion on alternate exchange rate regimes. First we shall learn about various exchange rate concepts.

Nominal Exchange Rates

The nominal exchange rate 'e' can be defined as the number of units of domestic currency (Rupee) required for purchasing one unit of foreign currency (dollar). For example, if Rs. 80 is required to buy one dollar, then the exchange rate $e = \text{Rs. 80 per dollar}$. Similarly, the exchange rate of the Indian rupee can also be defined in terms of other currencies, e.g., yen, euro, pound, etc.

When the exchange rate 'e' rises (say, from Rs. 80 to Rs.82 per dollar), there is an increase in the price of dollars in terms of rupees and there is **nominal depreciation** of the rupee vis-à-vis dollar (and **nominal appreciation** of the dollar vis-à-vis rupee). When 'e' falls (e.g., from Rs. 80 to Rs. 78 per dollar), the dollar becomes cheaper in terms of rupees and there is a **nominal appreciation** of the rupee (and **nominal depreciation** of the dollar vis-à-vis rupee).

Real Exchange Rate

While nominal exchange rates measure the rate of exchange between domestic and foreign currencies, the real exchange rate captures the rate at which domestic goods are exchanged for foreign goods. It is the price of foreign goods in terms of domestic goods and is computed using the nominal exchange rate and prices of a comparable basket of domestic and foreign goods as below:

$$r = (e p_f) / p_d \quad (4.6)$$

where 'e' is the nominal exchange rate (rupees per dollar), p_d denotes the domestic price level (in rupees) and p_f denotes the foreign price level (in dollars). ' $e p_f$ ' is the price in rupees of a foreign basket of goods and ' p_d ' is the price (in Rupees) of a comparable domestic basket.

The real exchange rate between rupee and dollar reflects the price of goods produced in the USA relative to those produced in India. A rise in r means American goods become relatively more expensive vis a vis Indian goods and there is **real depreciation** of the rupee; whereas a reduction in r makes Indian goods relatively more expensive and there is **real appreciation** of the rupee. The real exchange rate is often used as an index of a country's international price competitiveness. In case a country faces real appreciation of its currency, domestic goods become more expensive and this can lead to a reduction in demand for exports. In contrast, a real depreciation of the currency can boost export demand.

When domestic and foreign prices remain unchanged (that is, $\frac{p_f}{p_d}$ is constant), a nominal depreciation ('e' rises) of the domestic currency causes a corresponding real depreciation ('r' rises). In this case the nominal and real exchange rates

move in the same direction. However, prices may change simultaneously with exchange rates; e.g., an appreciation in nominal exchange rates ('e' falls), along with a large fall in domestic prices (p_d falls) may lead to a real depreciation (r rises). So, 'e' and 'r' may also move in opposite directions.

Exchange Rate Regimes

The exchange rate regime of a country determines how its exchange rate is determined.

Under a ***flexible (or floating) exchange rate regime***, the value of 'e' is determined by the demand and supply of foreign exchange, without any government intervention. In this case the currency is on a clean float or pure float. That means the exchange rate is fully flexible and adjusts continuously to equate market demand and supply of foreign exchange, so there is no role for intervention by the central bank and hence no need of foreign exchange reserves. In this case BOP balance is attained via exchange rate fluctuations. That is, in case there is a BOP surplus (supply of foreign currency exceeds demand), the exchange rate appreciates (e falls, so that given prices, r falls, so foreign goods are relatively cheaper), X falls, M rises and NX falls (the current account deficit widens and BOP surplus is reduced), till BOP equilibrium is restored. Floating exchange rates lead to considerable volatility in exchange rates that may be unfavorable for exporters and importers and may have adverse impacts on domestic production, investment and employment. As such, 'pure floats' are rarely observed in practice.

In most countries central banks maintain a stock of foreign exchange reserves and intervene and try to maintain the value of exchange rate within a certain band. Such exchange rate arrangements, widely prevalent across countries, are called a 'managed float' or dirty float. ***Intervention*** refers to the central bank buying or selling foreign exchange in an attempt to influence the exchange rate. In a managed float, the currency is allowed to float within a certain 'target zone' and the central bank stands ready to intervene (e.g., buy or sell dollars against rupees) whenever the exchange rate appears to breach the limits of this band.

At the other extreme we have the system of fixed exchange rate, wherein the exchange rate is not freely floating, but is fixed by the government. An important function of central banks in a fixed exchange rate system is to intervene in foreign exchange markets to keep the price of foreign exchange fixed at the pre-announced level. For this, central banks have to maintain adequate stock of foreign exchange reserves.

Under a fixed exchange rate system, the fixed parity announced by the central bank may be revised at times; in this case a decrease in 'e' due to official intervention is called ***revaluation*** while an increase in 'e' by official intervention is called ***devaluation***. In a world with few restrictions on international transactions in goods and assets, maintaining a fixed exchange rate poses an

immense challenge for a country's government. We will discuss a few aspects of policy dilemmas under a fixed exchange rate regime in the next section.

Check Your Progress 2

1) State whether the following statements are 'True' or 'False', giving reasons for your answer in each case:

- (a) A country can have a current account surplus, even when it has a negative trade balance (or merchandise trade deficit).
- (b) When Indian firms invest in foreign firms ('outward FDI'), there would be a corresponding capital inflow in the current account of India's BOP.
- (c) A country can have an overall BOP deficit even when it has a current account surplus.
- (d) When a country has a current account deficit (of say -\$20,000) and a capital account surplus (of say \$8000), its foreign exchange reserves would increase.

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2) Suppose that domestic currency depreciates by 2 percent and simultaneously domestic prices rise by 5 percent, while foreign prices remain unchanged.

- (i) What would be the impact on (a) real exchange rate; and (b) net exports?

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- (ii) How would your answer in (b) above would change if domestic prices remained constant following the nominal currency depreciation?

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3) Suppose a country experiences a sudden surge in capital inflows (e.g., FPI inflows) that create an excess supply of foreign exchange (say dollars). How would this affect the following?

- (a) the exchange rate and net exports under a flexible exchange rate regime

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- (b) the exchange rate and foreign exchange reserves under a fixed exchange rate regime ?

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4.4 LET US SUM UP

In this unit, we learnt that how openness to international trade and capital flows affects policy and macroeconomic outcomes within an economy. Our discussion on national income identities highlighted important differences between closed and open economies. For example, we saw that open economies can run current account deficits and borrow from the rest of the world to cover an excess of aggregate expenditure over income. Exchange rates are an important macroeconomic variable in open economies and from our analysis of BOP we got a flavor of the various factors, related to international trade in goods, services and assets that affect the demand and supply of foreign exchange and have an impact on nominal and real exchange rates. In this context we also learnt about alternate exchange rate regimes, viz., fixed and flexible exchange rate regimes.

4.5 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) Refer to Section 4.2 and answer.
- 2) Refer to Section 4.2 and answer

Check Your Progress 2

- 1) a) True
b) False
c) True
d) False
- 2) Refer to Sub-section 4.3.2 and answer.
- 3) Refer to Sub-section 4.3.2 and answer.

UNIT 5 OPEN ECONOMY

MACROECONOMICS -II*

Structure

- 5.0 Objectives
- 5.1 Introduction
- 5.2 The Open Economy IS-LM Framework
 - 5.2.1 The IS Curve
 - 5.2.2 The LM Curve
 - 5.2.3 Equilibrium in Open Economy IS-LM Model
- 5.3 Monetary and Fiscal Policies under Flexible Exchange Rate
 - 5.3.1 Fiscal Policy under Flexible Exchange Rate
 - 5.3.2 Monetary Policy under Flexible Exchange Rates
- 5.4 Monetary and Fiscal Policies under Fixed Exchange Rate
 - 5.4.1 Fiscal Policy under Fixed Exchange Rates
 - 5.4.2 Monetary Policy under Fixed Exchange Rates
- 5.5 Let Us Sum Up
- 5.6 Answer/Hints to Check Your Progress Exercises

5.0 OBJECTIVES

After going through this Unit you should be in a position to

- analyse short run macroeconomic equilibrium using the open economy IS-LM model; and
- discuss the limits to monetary and fiscal policies under fixed vis-à-vis flexible exchange rate regimes.

5.1 INTRODUCTION

In this Unit we will introduce the open economy IS-LM framework and use it to understand the complications involved in framing monetary and fiscal policy under alternate exchange rate regimes. In particular we consider two different exchange rate regimes, viz., fixed and flexible rate systems and explore the implications for conduct of monetary and fiscal policies under the assumption of unrestricted international capital mobility.

Apart from the above, we use the open economy IS-LM framework with perfect capital mobility to study the conduct of monetary and fiscal policy under alternate exchange rate regimes. First we examine the conduct of policy making under flexible exchange rates, followed by the case of a fixed exchange rate regime. This is known as the ‘Mundell-Fleming model’, named after economists Robert Mundell and Marcus Fleming, who developed this framework and analysed policymaking in open economies in the early 1960s.

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5.2 THE OPEN ECONOMY IS-LM FRAMEWORK

After our discussion above on certain basic concepts and accounting tools pertaining to open economies (see Unit 4), we will now examine output market equilibrium in the short run Keynesian framework using the open economy IS-LM model.

As in case of the closed economy IS-LM model, here also we assume that prices are given and there is 'excess capacity', so that output is demand determined.

5.2.1 The IS Curve

You are familiar with the IS curve in a closed economy (see Unit 3). In an open economy, aggregate demand (AD) is given by:

$$AD = C + I + G + X - M = C + I + G + NX, \quad (5.1)$$

where, $C = C(Y_d)$, is aggregate consumption expenditure which is a function of disposable income, $Y_d = Y - T$;

$I = I(i)$ is private investments, an inverse function of the rate of interest i ;

G represents exogenous government expenditure;

$X = X(Y_f, r)$ is exports, the demand for domestic goods by foreigners, which is a function of foreign GDP (Y_f) and the real exchange rate r ;

$M = M(Y, r)$ is imports, the demand for foreign goods by domestic agents, which is a function of domestic GDP (Y) and the real exchange rate r ;

$NX = (X - M) = NX(Y_f, Y, r)$ stands for the net exports that depend on the factors underlying demand for imports and exports, i.e., Y_f , Y and r .

Clearly imports are a leakage from the domestic economy as it is the expenditure of domestic agents on output that is produced in a different economy (hence part of their GDP) while exports are an addition to the total expenditure on domestic goods. Exports depend on Y_f , *ceteris paribus*, the higher is Y_f , higher would be export demand. Similarly, the higher is Y , higher would be the demand for imports. Both exports and imports depend on the real exchange rate which represents the relative price of foreign vis-à-vis domestic goods as discussed in equation (6). With an increase in r (a real depreciation), foreign goods become relatively more expensive. Thus, there is a decline in imports, while exports increase as domestic goods become relatively cheaper. So a rise in r would lead to a rise in X and a fall in M and net exports would be higher. Note that economists Alfred Marshall and Abba Lerner gave certain conditions under which a real devaluation or depreciation leads to an improvement in the trade balance (increase in NX), known as the *Marshall-Lerner condition*. According to the Marshall-Lerner condition, depreciation will lead to an improvement in the balance of trade of a country only if the sum of the price-elasticities of exports and imports of the country is greater than one. Throughout our discussion we assume that the Marshall-Lerner condition is satisfied.

Note that one of the assumptions in the IS-LM framework is that of given prices. Therefore, with given foreign (p^f) and domestic prices (p^d), a nominal depreciation or appreciation (rise or fall in e) brings about a real depreciation or appreciation (rise or fall in r).

As in the closed economy case, the IS curve, represents goods market equilibrium, so that along the IS curve we have aggregate demand equal to aggregate output or GDP (Y):

i.e., $Y = AD$

$$\text{or, } Y = C(Y_d) + I(i) + G + NX(Y_f, Y, r) \quad (5.2)$$

However, compared to the closed economy case, now there are two new determinants of aggregate demand, viz., foreign GDP and the real exchange rate. These two factors affect net exports and hence demand and output in an open economy. If Y_f increases (decreases) or r increases (decreases), NX would also increase (decrease) and the IS curve would shift to the right (left).

Another aspect of open economies reflected in (5.2) is that any change in aggregate demand (e.g., due to an increase in G), would affect Y and hence M and therefore lead to a change in NX . For instance, *ceteris paribus* an increase in government spending would lead to an increase in Y and M and thus reduce NX (i.e. reduce the size of the trade surplus or widen an existing trade deficit). However, if foreign GDP Y_f increases, *ceteris paribus*, exports would increase (as these are imports of the foreign country), also leading to an increase in Y , but in this case the trade balance would improve, i.e., NX would rise (assuming that rise in X due to increase in Y_f , would be more than the rise in M induced by rise in Y).

5.2.2 The LM Curve

So far as the LM curve is concerned, openness of an economy does not fundamentally change the inverse relation between money demand and the rate of interest (see Unit 3). If the demand for money is higher, the lower is the rate of interest, as in the case of a closed economy. Thus the LM curve, representing money market equilibrium (equality between demand for money and supply of money) is upward sloping. However, in an open economy, asset transactions with the rest of the world would give rise to capital flows (recorded in the capital account of the BOP). This also affects the rate of interest and money supply in an open economy. We discuss further on this issue below.

Let us assume that there is *perfect capital mobility*, i.e., there are no restrictions on international capital inflows and outflows and cross-border capital movements involve zero transaction costs. This is a simplifying assumption that captures the

reality of large-scale international capital mobility observed in the era of globalization, with advances in technology greatly reducing the cost of moving capital across international borders.

The assumption of perfect capital mobility has certain implications for the determination of interest rates in an open economy. It means that the domestic and foreign bonds are perfect substitutes, so that the rate of interest on domestic bonds (i) has to be in line with the rate of interest on foreign bonds (i_f), i.e., $i = i_f$ must hold. Here we assume that the foreign interest rate is given and not influenced by domestic economic policy.

If i_f is higher than i (i.e., $i_f > i$), foreign bonds would be preferred by investors owing to relatively higher returns on them. In this case, the domestic economy would experience capital outflows, as agents sell domestic bonds and buy dollars to purchase foreign bonds. The outflows continue until i rises (as price of domestic bonds fall) and equality is restored between i and i_f . When $i = i_f$, economic agents (such as households and firms) are indifferent between domestic bonds and foreign bonds.

Another point you should note regarding the money market in open economies relating to the supply of money and the role of foreign exchange reserves. Note that the stock of foreign exchange reserves is an asset of the central bank. When the central bank buys foreign currency, adding to its assets, it issues domestic currency so there is matching rise in its liability. As such, any change in foreign exchange reserves affects the monetary base or high powered money (liabilities of the central bank) and hence money supply in the economy, via the money multiplier.

Therefore, when a country experiences capital inflows and has a BOP surplus, this increases foreign exchange reserves and leads to an increase in money supply. Similarly, a capital outflow can result in a BOP deficit and a fall in money supply via a reduction in the stock of foreign exchange reserves.

In this context you should note that the central bank often uses a policy of **sterilization** to offset the impact of a change in foreign exchange reserves on the money supply. This can be done by using open market operations. In particular, central banks often use foreign exchange reserves for intervention in the foreign exchange market. Suppose the RBI intervenes to buy dollars (in order to absorb an excess supply of dollars) against rupees. This would add to its foreign exchange reserves and hence increase money supply. However, the RBI can carry out a sterilized intervention, by simultaneously selling bonds, in an open market operation, to reduce money supply. Thereby the RBI can either fully or partially

offset the impact of the increase in foreign exchange reserves on the money supply.

5.2.3 Equilibrium in Open Economy IS-LM Model

As in case of a closed economy, macroeconomic equilibrium occurs at the intersection of the IS and LM curves as shown in Fig. 5.1.

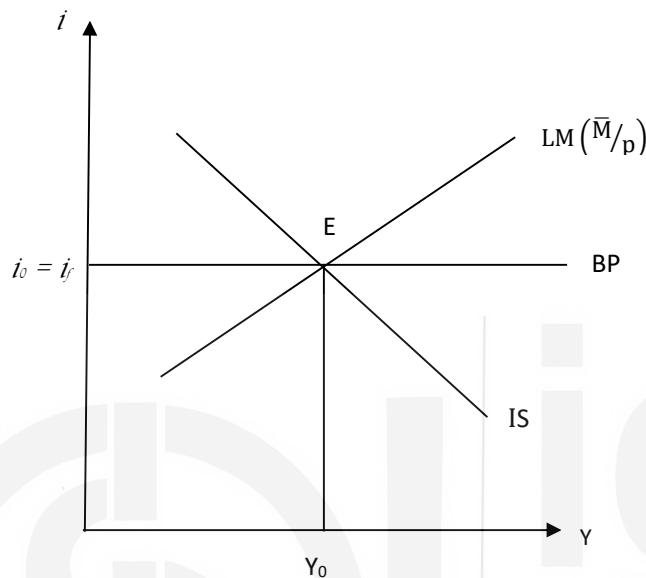


Fig. 5.1: “Macroeconomic Equilibrium in Open Economy IS-LM Model

Note that at the equilibrium E, the rate of interest, $i_0 = i_f$, owing to the assumption of perfect capital mobility. The point E represents macroeconomic equilibrium in the sense that there is both internal balance (goods and money market equilibrium) and external balance (BOP equilibrium).

The horizontal line BP that passes through the equilibrium rate of interest represents BOP equilibrium, in the sense that it represents interest rate – income combinations for which the current account surplus (deficit) is exactly offset by a capital account deficit (surplus). Note that here NX represents the current account balance.

With perfect capital mobility, the BP curve is horizontal, as the slightest deviation of i from i_f , would create BOP disequilibrium and trigger either infinite capital inflows or outflows. If $i > i_f$, there will be a surge in capital inflows and a BOP surplus, since inflows would be higher than required for BOP balance, at any given level of income. Whereas, or any $i < i_f$, there will be unlimited capital outflows and a BOP deficit. That is, for any Y (say Y_0), there is BOP surplus at all points above BP and a BOP deficit at all points below BP.

With imperfect capital mobility the BP curve would be upward sloping. As Y increases, M would rise, NX would fall and the current account deficit would widen. To maintain BOP balance therefore, i would have to increase to attract capital inflows and create a matching capital account surplus. With imperfect